MILITARY SPECIFICATION

TEST HARNESS, RADAR SET AN/APM-95A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The test equipment covered by this specification shall be for shop use and shall facilitate the testing of Radar Set AN/APG-53A or -53B.

1.2 Classification - The test equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Para.
Radar Set Control	C-2450/APG-53A	3.5.1
Distribution Box	J-2631/APM-95A	3.5.2
Maintenance Fixture, Radar Set	MT-2067/APM-95	3.5.3
Maintenance Fixture, Computer and Power Supply	MT-2064/APM-95	3.5.3
Maintenance Fixture, Indicator	MT-2066/APM-95	3.5.3
Maintenance Fixture Control	MT-2065/APM-95	3.5.3
Maintenance Fixture, Power Supply - Indicator	MT-3370/APM-95A	3.5.3
Cable Assembly	W6 of AN/APM-95	3.5.4
Cable Assembly	W7 of AN/APM-95	3.5.4
Cable Assembly	W8 of AN/APM-95	3.5.4
Cable Assembly	W9 of AN/APM-95	3.5.4
Cable Assembly	W10 of AN/APM-95	3.5.4
Cable Assembly	W11 of AN/APM-95	3.5.4
Cable Assembly, Special Purpose, Electrical	W12 of AN/APM-95A	3.5.4
Cable Assembly, Special Purpose, Electrical, Branched	W13 of AN/APM-95A	3.5.4
Extractor Assembly, Module	64A29D39	3.5.5

1.3 Associated Equipment - The test harness shall operate with the associated equipment listed in 6.8.

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitations for bids form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-T-5422	Testing Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-15362	Harness; Test Bench, Avionic Equipment General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of

LIST OF DRAWINGS

Naval Air Systems Command

Drawings on List of Drawings LD 64A29B1 for Test Bench Harness AN/APM-95A (refer to 6.9)

Drawings on List of Drawings 57A121K1 for Radar Set Control C-2450/APG-53A

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

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3.2 Parts and Materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern.

(1) Parts and materials requirements shall conform to MIL-T-21200.

(2) Nonrepairable subassemblies as outlined in MIL-T-21200, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-21200.

3.3 Design and Construction - The test harness shall be manufactured in accordance with drawings and subsidiary specifications listed in LD 64A29B1 and related lists insofar as compatible with the requirements of this and related specifications. It is a condition of final acceptance that the test harness manufactured meet all the performance requirements of this specification and such acceptance test documents as may be applicable. Proposed changes, either to correct errors in the design disclosure or to effect improvements, shall be specifically approved in writing by the procuring activity unless otherwise specified.

3.3.1 Total Weight - The total weight of the test harness including cables shall be a minimum consistent with good design and shall not exceed 129 pounds.

3.3.2 Reliability -

3.3.2.1 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.3 Cables and Connectors - The test harness shall provide for the use of cables and connectors in accordance with MIL-T-21200.

3.3.4 Interchangeability - The test harness shall meet the interchangeability requirements as defined in MIL-T-21200. The drawings listed on LD 64A29B1 shall be the standard for measuring interchangeability of all applicable assemblies, subassemblies, and parts.

3.3.5 Interference Control - The generation of radio interference by the test harness and the vulnerability of the test harness to radio interference shall be controlled within the limits of MIL-I-6181.

3.3.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field test and maintenance.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

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3.3.7.1 Cable Marking - The marking of cables shall be in accordance with MIL-N-18307, except the labeling sleeves shall be a polyvinyl-chloride synthetic insulation in accordance with MIL-I-631 (Type F, Forms, Grade A, Class 1, Category I) in lieu of metal bands.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30 ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
*Input power voltage	115 ± 1.0 V, 400 cps AC, ±5 cps, 1 phase 6.0 ± 0.3 V, 400 cps AC, ±5 cps 27 ± 0.5 V DC 250 ± 2.0 V DC

*Note: Input power voltage listed is voltage used when the test harness is interconnected with the radar set and activated.

3.3.9 Service Conditions - The test equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-H-15362, except that the temperature and salt spray conditions for the extractor module assembly shall be as follows:

3.3.9.1 Temperature - The module extractor assembly shall meet the temperature conditions of MIL-STD-202 method 102 test condition D except temperature extremes shall be -30°C (+0 or -3°C) and +65°C (+3 or -0°C).

3.3.9.2 Salt Spray - The module extractor assembly shall meet the salt spray conditions of MIL-STD-202 method 101 test condition B (48 hours).

3.3.10 Primary Input Power Requirements - The test harness shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-21200. The power required shall be of the following types and shall not exceed the specified amounts.

AC Power, Single Phase, 115 V, 380-420 cps, 20 VA

AC Power, Single Phase, 6.0 V, 380-420 cps, 8 VA

DC Power, 27.5 V, 7 VA

DC Power, 250 V, 0.25 VA

3.3.10.1 Protection - The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as defined in MIL-STD-704.

3.3.11 Insulation Resistance - The electrical circuits of the test harness shall have an insulation resistance of at least five (5) megohms at 500 volts rms, 60 cycles, when measured between the housing and each isolated circuit, and between each and every pair of isolated circuits.

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3.3.12 Dielectric Strength - All electrical circuits in the test harness which are ungrounded and required to be insulated from each other shall be capable of withstanding 500 volts rms, 60 cycles at room temperature and humidity for a period of ten (10) seconds when this voltage is connected between each pair of isolated circuits, and between each isolated circuit and the housing.

3.4 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation will be specified.

3.5 Detail Requirements -

3.5.1 Radar Set Control C-2450/APG-53A - The radar set control shall be manufactured in accordance with the documents listed on NASC list of drawings 57A121ASK1 and shall meet the following requirements:

3.5.1.1 Function - The radar set control shall be used in the test harness to cycle the radar set through its various modes of operation.

3.5.1.2 Form Factor - The maximum dimensions of the radar set control shall be 6-1/8 inches long by 5-3/4 inches wide by 8-1/4 inches high.

3.5.1.3 Weight - The weight of the radar set control shall not exceed 4.0 pounds.

3.5.1.4 Controls - The radar set control shall contain the following controls:

3.5.1.4.1 ANGLE OF ATTACK ON-OFF Switch - The ANGLE OF ATTACK ON-OFF switch shall be a single pole switch used in switching the angle of attack function into or out of the radar set.

3.5.1.4.2 SCAN SHORT-LONG Switch - The SCAN SHORT-LONG switch shall be a double-pole, double-throw switch which permits the selection of long or short range when in the plan, profile, or search mode of operation.

3.5.1.4.3 ANTENNA ELEVATION Control - The ANTENNA ELEVATION control shall be a transmitter synchro that permits control of the antenna elevation angle. When the control is varied from +10 to -15° the output across stator connections S1, S2, and S3 shall indicate the angular change.

3.5.1.4.4 MODE SELECTOR Control - The MODE SELECTOR control shall be a 7-position, 6-deck switch that permits the selection of various modes of radar set operation.

3.5.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J7001	MS3102A28-21P	Radar Set Inputs

3.5.2 Distribution Box J-2531/APM-95A - The distribution box shall be manufactured in accordance with NASC drawing 64A29D6 and shall meet the following requirements:

3.5.2.1 Function - The distribution box provides test points for the testing of radar set parameters.

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3.5.2.2 Form Factor - The maximum dimensions of the distribution box shall be 8-15/16 inches long by 2-1/8 inches wide by 4-3/16 inches high.

3.5.2.3 Weight - The weight of the distribution box shall not exceed 2.0 pounds.

3.5.2.4 Controls - The distribution box shall contain the following controls:

3.5.2.4.1 APR 23-APG-53B RADAR Switch - The APR 23-APG-53B RADAR switch shall be a single-pole toggle switch that shall open or close the APG-53B UNBLANKING TRIGGER, and VIDEO circuits used in conjunction with Passive Radar Homing System AN/APR-23.

3.5.2.4.2 SHRIKE-APG-53B RADAR Switch - The SHRIKE-APG-53B RADAR switch shall be a single-pole toggle switch that shall open or close the APG-53B AZ-1CPS and EL-1CPS circuits used in conjunction with SHRIKE missiles.

3.5.2.4.3 TV MODE-OFF-RADAR Switch - The TV MODE-OFF-RADAR switch shall be a double-pole, double-throw toggle switch that determines whether the radar set indicator is in the radar or television mode of operation when the switch is placed to the RADAR or TV MODE position.

3.5.2.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J102	MS3112E14-19P	PP-4247/APG-53B Inputs
J103	MS3112E22-55DW	IP-774/APG-53B Inputs
J104	MS27035-625B/U	APR-23 Unblanking
J105	MS27035-625B/U	Radar Trigger
J106	MS27035-625B/U	Radar Video

3.5.3 Maintenance Fixtures - The test harness shall contain the maintenance fixtures as listed in table I and shall be manufactured in accordance with the applicable NASC drawing.

3.5.3.1 Circuits - The transformer circuit of Maintenance Fixture, Control MT-2065/APM-95 shall be 5.5 ± 0.5 volts when 115 ± 1 volts, 400 ± 5 cps is applied to the transformer primary connector pins.

3.5.3.2 Controls - Maintenance Fixture, Control MT-2065/APM-95 shall contain the following controls:

3.5.3.2.1 ALTITUDE COMPUTER - WITH DIVE TOSS COMPUTER Switch - The ALTITUDE COMPUTER - WITH DIVE TOSS COMPUTER switch shall be a toggle switch which opens or closes the radar set transmitting circuit.

3.5.3.3 Electrical Connections - Connections to external circuits for Maintenance Fixture, Control MT-2065/APM-95 shall be provided as follows:

Reference Designation	Receptacle Type	Function
J7002	58A5A126-12-10P	Radar Set Inputs

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Table I

Test Harness Fixtures

Qty	Fixture	Mfg. NASC Drawing	Max. Over-All Size (inches)			Max. Wt (lbs)
			L	W	H	
1	Maintenance Fixture, Radar Set MT-2067/APM-95	58A6F25	21-5/8	15-1/16	25-7/16	31
2	Maintenance Fixture, Computer and Power Supply MT-2064/APM-95	58A6F65	18	10-1/2	7-7/8	16
1	Maintenance Fixture, Indicator MT-2066/APM-95	58A6F85	18-1/2	10-1/2	12-1/8	19
1	Maintenance Fixture, Control MT-2065/APM-95	58A6F105	6-1/4	10-1/2	9-5/8	3
1	Maintenance Fixture, Power Supply - Indicator MT-3370/APM-95A	64A29E18	19-5/8	10-1/2	10	18

Connections to external circuits for Maintenance Fixture, Radar Set MT-2067/APM-95 shall be provided as follows:

Reference Designation	Receptacle Type	Function
J326	58A5A89-1	Radar Set Connection
J328	58A5A89-1	Radar Set Connection
J329	58A5A89-1	Radar Set Connection
J330	58A5A89-1	Radar Set Connection

3.5.4 Cable Assemblies - The test harness shall contain the following cable assemblies which shall be manufactured in accordance with the applicable NASC drawing:

Cable Assembly	Mfg NASC Drawing
W6	58A6C8
W7	58A6C9
W8	58A6C10
W9	58A6C11
W10	58A6C12
W11	58A6F13
W12	64A29D27
W13	64A29F29

3.5.5 Extractor Assembly, Module - The module extractor assembly shall be manufactured in accordance with NASC drawing 64A29D39 and shall meet the following requirements:

3.5.5.1 Function - The module extractor assembly is used to disconnect modules

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from the chassis of Azimuth-Elevation-Range Indicator IP-868/APA-167A.

3.5.5.2 Form Factor - The maximum dimensions of the module extractor assembly shall be 0.645 inches long by 5.54 inches wide by 5.11 inches high.

3.5.5.3 Weight - The weight of the module extractor assembly shall not exceed 0.5 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-H-15362 and interference tests in accordance with MIL-I-6181 with the exception that the temperature and salt spray tests for the module extractor assembly shall be in accordance with MIL-STD-202.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

Nonrepairable subassemblies must be Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:29Z
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(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

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*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first ten produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational tests at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be

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performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, or Special Tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient (30 ± 10°C)
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room Ambient
Input Voltage	115 ± 1.0 volts (at lowest applicable frequency) 6.0 ± 0.3 volts (at lowest applicable frequency) 27.0 ± 0.5 volts DC 250 ± 2.0 volts DC
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion

fixtures as listed in table I above may be

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of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555.

6. NOTES

6.1 Intended Use - The test harness described by this specification shall be used to facilitate the testing of Radar Set AN/APG-53A or -53B.

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltages of 27 ± 0.5 V DC, 250 ± 2.0 V DC, 6.0 ± 0.3 V AC, and 115 ± 1.0 V AC as applicable (see 3.3.8). Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages. These characteristics

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shall be used when testing the equipment for compliance with the minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions, the nonrepairable subassembly should evidence a mean-time-to-failure greater than 5,000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Selection of applicable levels of packaging and packing (see 5.1).

6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Test Set, Weapon Control Aircraft AN/ASM-184(XN-1)

(2) Test Set, Radar TS-966/APG-53 with associated cables W1, W2, W3, W4, W5, and W12.

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6.9 Applicable Referenced Drawings - All drawings on LD 64A29B1 and 57A121ASK1 are applicable drawings for the test harness. The following drawings are referenced in the Detail Requirements (3.5) and are listed below for convenience:

MS27035-625B/U	Connector, Receptacle, Electrical, Series BNC, UG-625B/U
MS3102A28-21P	Connector
MS3112E14-19P	Connector
MS3112E22-55DW	Connector
58A5A126-12-10P	Connector, Receptacle Electrical
58A5A89-1	Adapter, Connector, Bulkhead, Hermetic Seal
58A6C10	Cable Assembly W8 of AN/APM-95
58A6C11	Cable Assembly W9 of AN/APM-95
58A6C12	Cable Assembly W10 of AN/APM-95
58A6C8	Cable Assembly W6 of AN/APM-95
58A6C9	Cable Assembly W7 of AN/APM-95
58A6F105	Maintenance Fixture, Control MT-2065/APM-95
58A6F13	Cable Assembly W11 of AN/APM-95
58A6F25	Maintenance Fixture, Radar Set MT-2067/APM-95
58A6F65	Maintenance Fixture, Computer and Power Supply MT-2064/APM-95
58A6F85	Maintenance Fixture, Indicator MT-2066/APM-95
64A29D27	Cable Assembly, Special Purpose, Electrical W12 of AN/APM-95A
64A29D39	Module Extractor Assembly
64A29D6	Distribution Box J-2531/APM-95A
64A29E18	Maintenance Fixture, Power Supply-Indicator MT-3370/APM-95A
64A29F29	Cable Assembly, Special Purpose, Electrical, Branched W13 of AN/APM-95A

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-T-21390A(AS) TEST HARNESS, RADAR SET AN/APM-95A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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